The tectonic significance of some basic dyke swarms in the Canadian Superior Province with special reference to the geochemistry and paleomagnetism of the Mistassini swarm, Quebec, Canada

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The Mistassini dykes extend northwest from the Mistassini embayment and comprise both tholeiitic and komatiitic suites. They are probably >2000 Ma old and yield two major paleomagnetic components. One of these, with a pole at 131°W, 13°S, is thought to be an overprint related to the Elsonian Disturbance 1400–1500 Ma ago. A very steeply down (and reversed) component may be primary and has a pole at 080°W, 50°N.

These spacial, chemical, and age relationships between the Mistassini, Molson, Marathon, and Payne River dyke swarms and the Aphebian supracrustal fold belts on the perimeter of the Superior Province suggest a genetic relationship between the dyke swarms and the fold belts. The supracrustal belts are evidence of the opening and closing of oceans, and the dyke swarms are evidence of early-stage failed arms related to these openings. More rarely (for example, the Payne River dykes), early-stage dyke swarms are developed and preserved parallel to the edges of newly developed spreading plate boundaries. Presumably if a spreading episode stops, the dykes themselves may remain as the only evidence of that event. Probably all the world's great continental dyke swarms have the above-described tectonic setting, and the number and extent of dyke swarms during a geological epoch may be a measure of the number and vigour of spreading events.

Les dykes de Mistassini s'étendent au nord-ouest de la baie de Miatassini, et ils sont formés de séquences tholéiitiques et komatiitiques. Leurs âges remontent probablement à >2000 Ma, et ils fournissent deux composantes palaléomagnétiques majeures. Celle dont le pôle est 131°O, 13°S est considérée comme étant une surimpression due à l'orogenèse de l'Elsonien, il y a 1400–1500 Ma. La composante de forte plongée (et inverse) dont la pôle est à 080°O, 50°N pourrait bien être primaire.

Les relations spatiales, chimiques et géochronologiques entre les réseaux de dykes de Mistassini, Molson, Marathon et de la rivière Payne avec les zones de plissements supracrustales aphébiennes sur la périmètre de la province du lac Supérieur révèlent l'existence d'un lien génétique entre les réseaux de dykes et les zones de plissements. Les zones supracrustales témoignent de l'ouverture et de la fermeture des océans, et les réseaux de dykes indiquent un stade précoce de bras fracturés associés aux ouvertures océaniques. Les réseaux de dykes du stade précoce (par exemple, celui de la rivière Payne), avec une orientation parallèle aux bordures de la crête d'expansion océanique, sont plus rarement développés et préservés. Dans le cas d'arrêts du stade d'expansion océanique, les dykes seuls pourraient constituer une preuve suffisante d'un tel événement. Il est probable que tous les grands réseaux de dykes continentaux aient un contexte tectonique identique à celui que nous décrivons, et le nombre ainsi que l'étendue des réseaux de dykes durant une époque géologique pourraient refléter le nombre et l'intensité des événements d'expansion océanique.

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Introduction

One purpose of this paper is to discuss the petrochemistry, paleomagnetism, and global tectonic significance of a swarm of basic dykes that occurs in the Superior Province adjacent to the central Grenville Front. The major concentration of these dykes trends northwest (Fig. 1) from the Mistassini–Otish embayments, so much so as to suggest a genetic relationship between the occurrence of these embayments and intrusion of the dyke swarm. Some of the dykes show up as spectacular linear highs on regional aeromagnetic maps (e.g., Geological Survey of Canada maps 7112G and 7115G). Regional mapping north of the 52nd parallel (Eade 1966) indicates that minor similar dykes of this trend extend almost to Hudson Bay.

The dykes form two chemical suites, one of which is komatiitic, the other high-iron tholeiite to tholeiitic andesite. The dykes, however, are parallel and contain a common paleomagnetic component. We later refer to the relationship between tholeiitic and komatiitic dykes of the Molson swarm, which are related to Fox River volcanics of similar age and composition. The evidence favours the idea that the two magma types can be generated essentially simultaneously. We think this is the case with the Mistassini swarm.

Most of the samples used in the petrochemical study were collected by E.H.C., D.J., and others during mapping of the Wabissinane River area (Chown and Caty 1972), and chemical analyses of these samples were carried out by D.J. and N.M. at the Geochemistry Laboratory of the University of Montreal (Table 1).

TABLE 1. University of Montreal chemical analyses of Mistassini dykes (sample locations in Fig. 1)

	DJ3001	DF1801	DF2001	DF2401	DJ1308	DJ1309	DJ2722	DJ4205	EC0710	MW0603	MW0810	MW1204	MW1303
SiO_2	48.7	49.0	49.6	50.0	51.2	50.5	52.1	47.5	50.5	51.7	54.9	48.3	49.2
Al_2O_3	2.02	4.1	15.7	16.4	13.3	12.1	17.0	11.2	11.4	13.6	16.4	13.0	11.9
TiO_2	0.04	0.13	2.68	2.80	0.84	1.0	1.0	0.78	0.79	0.53	0.66	1.59	0.80
Fe_2O_3	6.4	12.6	14.6	13.7	11.4	12.0	9.8	12.2	11.4	10.9	7.6	14.7	12.8
MnO	0.10	0.20	0.17	0.23	0.16	0.19	0.12	0.19	0.18	0.15	0.15	0.20	0.20
MgO	42.4	30.3	4.09	3.74	10.0	13.3	6.53	16.03	13.7	11.0	7.12	12.1	14.8
CaO	0.10	3.05	6.10	6.43	8.77	7.95	7.94	8.67	8.69	8 74	7.88	6.35	8.06
Na_2O	0.20	0.42	4.1	4.8	2.4	2.2	3.4	2.7	1.8	2.3	3.4	3.3	1.8
K_2O	0.012	0.130	2.38	1.18	1.94	0.80	2.02	0.61	1.36	1.02	1.88	0.19	0.43
P_2O_5	0.01	0.02	0.62	0.67	0.08	0.11	0.11	0.08	0.08	0.07	0.09	0.17	0.09
Cr	3700	2600	45.2	45.9	698	822	401	1367	1550	639	369	570	1070
Ni	2660	980	47.3	51.9	598	638	101	808	629	358	191	448	589
Zn	49.2	0.0	0.0	0.0	0.0	0.0	114	75.0	74.6	64.9	59.1	164	0.0
Rb	0.0	10.3	40.2	44.8	131	47.0	54.3	33.3	60.8	34.1	65.2	16.8	20.8
Sr	0.0	7.4	512	287	262	94.6	421	93.4	157	123	355	202	108
Ba	0.0	40.0	798	530	230	185	0.0	0 0	0.0	0.0	0.0	0.0	90.6
Pb	6.2	0.0	5.5	4.5	5.8	1.2	8.1	2.0	4.6	2.5	3.2	2.2	1.4
Th	1.8	0.0	0.0	0.9	3.4	3.4	0.6	1.2	2.6	2.0	3.0	1.0	3.6

	MW1401	MW2108	TC0604	TC1112	TC1120	CB1711	CB2508	DF2303	DF2312	DF2606	DF3901	DJ0602	DJ1112
SiO_2	48.1	50.2	49.0	48.5	51.1	54.1	55.6	53.2	59.0	56.8	55.6	59.2	54.2
Al_2O_3	11.4	11.2	10.2	15.4	12.7	13.7	14.4	13.3	13.9	14.2	16.3	13.3	14.00
TiO_2	0.78	0.83	0.97	2.73	0.71	2.53	2.29	2.44	1.82	2.31	2.13	1.96	2.53
Fe_2O_3	12.1	11.6	13.6	14.9	11.0	13.9	12.8	14.4	12.3	12.4	10.1	11.8	13.7
MnO	0.18	0.18	0.20	0.18	0.17	0.19	0.24	0.19	0.17	0.18	0.18	0.15	0.19
MgO	16.8	14.3	15.6	4.56	12.1	3.04	2.71	3.50	2.18	2.78	2.86	2.14	2.85
CaO	7.77	8.80	7.97	7.41	8.97	5.60	4.91	6.29	3.99	4.74	5.22	3.40	5.61
Na_2O	2.4	1.4	1.6	3.6	1.5	3.6	4.1	3.6	3.4	3.4	4.1	3.5	3.6
K_2O	0.39	1.33	0.80	1.92	1.60	2.46	2.30	2.48	2.78	2.44	2.80	3.79	2.39
P_2O_5	0.08	0.08	0.10	0.71	0.09	0.84	0.69	0.65	0.52	0.69	0.71	0.71	0.93
Cr	1560	1000	978	49.0	1300	33.1	36.7	19.0	39.7	37.4	22.0	29.9	30.3
Ni	743	735	971	53.1	590	21.5	18.3	25.1	7.0	16.4	21.6	43.2	18.6
Zn	109	89.3	0.0	0.0	0.0	0.0	0.0	127	0.0	0.0	110	0.0	0.0
Rb	18.1	58.7	56.3	46.8	91.6	43.9	42.1	46.3	58.6	44.1	60.5	79.4	52.7
Sr	96.3	213	113	467	237	365	453	397	435	342	407	289	346
Ba	0.0	0.0	113	647	223	1080	1180	0.0	901	1070	0.0	1052	808
Pb	4.3	2.6	4.4	4.8	2.8	6.3	6.4	10.9	10.3	3.7	5.6	99.5	4.5
Th	0.9	2.7	4.3	1.4	2.1	0.7	0.0	5.0	0.0	0.0	0.7	0.0	1.6

	DJ2002	DJ2212	DJ2708	DJ2710	DJ3414	DJ3902	JL0301	JL2302	TC1902	TC2006	TC2705	TC2805	MW1101
SiO_2	53.6	54.4	57.2	52.3	53.8	52.0	52.3	52.6	50.6	53.5	53.7	53.6	53.7
Al_2O_3	14.2	13.8	15.1	13.7	14.1	13.8	13.8	13.8	13.6	14.9	14.0	13.8	14.0
TiO_2	2.39	2.19	2.05	1.52	2.60	2.68	2.59	2.37	2.64	2.58	2.37	2.57	2.52
Fe_2O_3	14.1	13.0	13.2	12.0	13.6	13.8	14.0	14.0	15.4	14.4	13.3	14.0	13.9
MnO	0.19	0.18	0.11	0.16	0.16	0.20	0.16	0.18	0.21	0.18	0.18	0.22	0.20
MgO	3.19	2.86	4.09	5.70	3.51	3.42	3.24	3.37	3.08	3.41	3.42	3.21	2.81
CaO	5.74	5.65	3.64	8.84	5.57	5.42	6.30	6.39	6.27	3.86	6.04	5.43	5.68
Na_2O	3.7	4.1	2.8	3.0	4.1	4.5	4.3	4.0	4.6	3.9	3.9	4.0	3.6
K_2O	2.56	2.67	2.60	1.23	2.17	3.03	2.30	2.54	2.77	2.53	2.47	2.34	2.55
P_2O_5	0.72	0.75	0.48	0.17	0.80	0.84	0.95	0.67	0.81	0.70	0.64	0.87	0.97
Cr	34.7	37.5	45.8	228	24.4	18.7	27.0	27.5	29.9	21.7	42.3	30.1	32.1
Ni	20.1	14.5	21.3	72.7	22.1	23.0	16.5	25.4	19.3	24.9	30.1	13.9	17.9
Zn	0.0	0.0	222	109	120	135	0.0	105	124	142	105	0.0	0.0
Rb	53.9	67.6	105	35.4	33.5	46.1	43.2	38.6	41.5	31.8	41.7	39.7	46.2
Sr	443	464	289	401	221	234	224	456	358	256	394	340	354
Ba	901	975	0.0	0.0	0.0	0.0	717	0.0	0.0	0.0	0.0	1010	887
Pb	2.8	6.1	10.7	12.1	5.3	9.5	8.8	8.6	6.2	3.6	0.8	4.5	7.1
Th	2.3	1.8	0.3	5.1	0.0	0.0	0.5	0.3	2.0	0.0	2.0	0.8	1.2

NOTES: Major- and trace-element concentrations were determined by techniques described by Gunn (1967) and Watkins *et al.* (1970) using United States Geological Survey BCR and W-1 as calibration standards. Total iron is reported as Fe_2O_3. The analytical results are not strictly comparable with those of Table 2 because CO_2 and water are not reported in the above analyses.

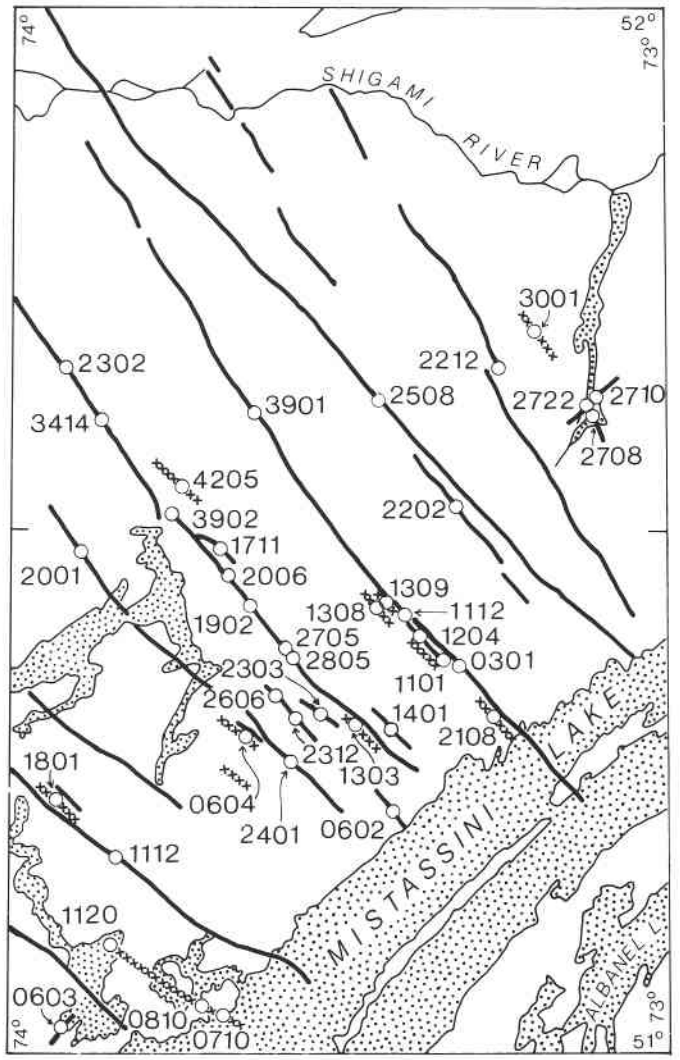


FIG. 1. Distribution of major Mistassini dykes and location of samples chemically analysed at the University of Montreal (Table 1). Prefixes of the sample numbers have been omitted from the figure. Dykes shown as rows of ×'s are komatiitic; those shown as solid lines are tholeiitic.

The samples used for the paleomagnetic study (114 separately oriented cores and blocks from 20 sites) were collected by W.F.F. in 1965 and by W.F.F. and E.H.C. in 1976 (Fig. 2). Except for site 7603, chemical analysis of at least one sample from each of the 20 paleomagnetic collecting sites was carried out in the analytical laboratories of the Geological Survey of Canada (Table 2).

Field relationships

The dykes range in size from discontinuous lenses less than 1 m thick to sheets 100 m thick and up to 150 km long. The main dyke pattern, which is derived largely from aeromagnetic maps, is shown in Fig. 1. All the dykes are close to vertical, several bifurcate, and a few have subrounded to angular apophyses 300–500 m wide.

Dyke material is present only as a few discontinuous dykes at the bedrock surface of one of the most prominent linear magnetic anomalies (see 2001, Fig. 1). However, all the other linear anomalies have at least one continuous dyke as a surface expression. Commonly, a major continuous dyke has one or more minor parallel dykes associated with it. In some cases a minor dyke can be traced into the large dyke, but most small parallel dykes do not appear to be connected to their nearby major dyke and are of different composition.

The Mistassini dykes intrude a northeast-trending amphibolitic–quartzo-feldspathic gneiss–granodiorite terrain of Archean age. Contacts are sharp, with very few inclusions of intruded rock. Most of the dykes die out near the northwest shore of Mistassini Lake. This coincides with the first appearance of northeast-trending strata of the Mistassini Group (Chown and Caty 1973). The dykes have never been observed to cut the Mistassini Group, but the aeromagnetic anomalies caused by some dykes extend a short distance into the area underlain by these strata. The gradual fading of the anomalies suggests that the dykes are overlain unconformably by the Mistassini Group. One northeast-trending dyke cuts the Papaskwasati Formation (Chown 1971), a clastic equivalent of the carbonate members of the middle Mistassini Group (Chown and Caty 1973). A small, sill-like mass of gabbro cuts dolomite in the centre of the Mistassini Basin (Neilson 1966). It is not known whether the northeast dyke and the gabbro sill are genetically related to the northwest-trending Mistassini dykes. Less prominent, northwest-trending, presumably Mistassini dykes have been mapped to the northeast as far as the Otish Mountains, where tholeiitic sills and feeder dykes cut sandstones of the Otish Group (Fahrig and Chown 1973).

Age

Radioisotopic dating of the Mistassini dykes is not entirely satisfactory. A whole-rock K–Ar age of 1925 ± 60 Ma (Wanless *et al.* 1972, pp. 60–61) was obtained from chilled marginal dyke material at site 6556 (Fig. 2), and at this contact biotite from immediately adjacent country rock yielded an age of 1960 ± 58 Ma (Wanless *et al.* 1968, pp. 112–113). Experience with K–Ar analysis of chilled, fresh diabase and biotite from within dykes or from contact country rock indicates that these materials generally give ages 10–15% lower than those obtained by Rb–Sr or U–Pb methods. These determinations therefore suggest an age of intrusion of at least 2000 Ma, probably as great as 2200 Ma. On the other hand, chilled material from a dyke (site 6548, Fig. 2) of similar but slightly more andesitic composition yielded a whole-rock K–Ar age of 1220 ± 45 (Wanless *et al.* 1972, p. 60). Another chilled sample from a tholeiitic dyke of Mistassini trend but a considerable distance to the northwest yielded a whole-rock K–Ar age of 1265 ± 165 Ma (Wanless *et al.* 1970, p. 67). We feel that the good concordance between the chilled whole-rock and contact biotite ages of site 6556 makes these more reliable estimates of the age of intrusion than the chilled whole-rock age (1220 Ma) of site 6548. The lesser age may reflect overprinting.

Petrography

Tholeiitic suite

Dykes of this suite form the thicker and longer representatives of the swarm and are responsible for the major aeromagnetic anomalies. No cross-cutting relationships are known between dykes of this suite and those of the later-described komatiitic suite. In several of the dykes plagioclase phenocrysts with long dimensions up to 6 cm form striking layers. The phenocrysts are slightly zoned and partly saussuritized. The remainder of the dyke material, whether or not phenocrysts are present, consists chiefly of slightly zoned

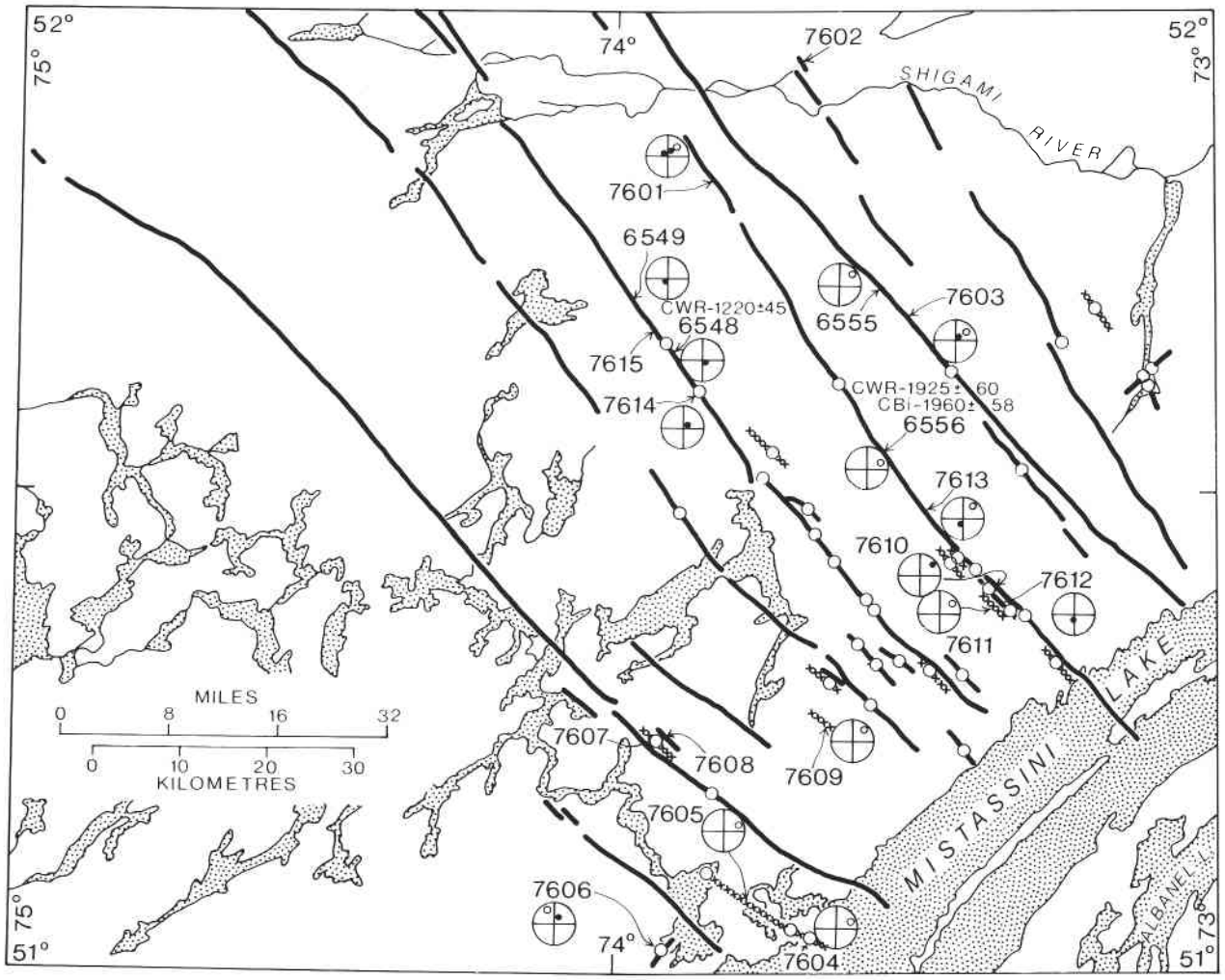


FIG. 2. Location of paleomagnetic sampling sites (at ends of arrows). The large circles are diagrammatic stereonets showing the dominant magnetization direction. Material for Geological Survey of Canada chemical analyses (Table 2) was also obtained from these sites. Small unlabelled circles are locations of samples analysed at the University of Montreal (Table 1). Locations of samples used for K−Ar age determination are shown: CWR = chilled whole-rock; CBi = country-rock contact biotite.

plagioclase (An_{40-35}) intergrown with stubby, pale brown clinopyroxene. The groundmass plagioclase is fresh, but the clinopyroxene is extensively uralitized. Quartz and a micrographophyric intergrowth of quartz and potash feldspar occur interstitially and are usually intergrown with spongy outer rims of plagioclase laths. Minor constituents include 1−2% skeletal opaque iron ores, commonly rimmed by brown−green biotite and 1−2% apatite.

Komatiitic suite

The suite ranges from ultramafic komatiite through basaltic komatiite into the high-magnesium tholeiitic basalt field. The ultramafic rocks are almost completely altered. Olivine is converted to masses of serpentinite with sagenitic webs of dusty magnetite, which outline rounded grain shapes and the fracture pattern of the original olivine. Interspersed with, and enveloping these, are bastite grains pseudomorphing the original large pyroxene crystals. Veins of serpentinite cut the massive parts of the rock.

One dyke is a completely uralitized pyroxenite in which pale green amphibole preserves the pyroxene grain outlines. Dusty opaque iron ores coat amphibole cleavages, and minor flakes of phlogopite are interspersed among the grains.

The komatiitic basalts are composed of plagioclase and pale pink clinopyroxene in ophitic or subophitic intergrowths with 5−10% olivine, which is generally enclosed in the pyroxene. Plagioclase is strongly zoned from An_{50} to An_{26}. What appears to be a patchy deuteric alteration converts the plagioclase to a mass of saussurite and sericite. The pale pink clinopyroxene in these alteration patches is only partly uralitized. Olivine is largely converted to serpentinite, talc, or colourless amphibole, leaving rare relict grains. Apatite and opaque ores constitute up to 5% of the rock. The iron ores are surrounded by a thin jacket of red biotite and are converted to leucoxene in more altered rocks.

Chemistry

On a Jensen plot (Jensen 1976), the Mistassini dykes appear to form two distinct chemical suites (Fig. 3). One falls in the tholeiitic andesite and high-iron tholeiitic basalt fields; the other follows a komatiitic trend from ultramafic komatiite into high-magnesium tholeiitic basalt, with the greatest concentration in the basaltic komatiitic field. The apparent gap between the two suites may be a real one. The 13 komatiitic samples analysed at the University of Montreal have MgO contents

TABLE 2. Geological Survey of Canada chemical analyses of Mistassini dykes (sample locations in Fig. 2)

	760102	760202*	760408	760505	760602	760603	760701	760803	760906	761003	761008*	761101	761202
SiO_2	53.3	49.6	49.1	48.9	50.7	50.5	42.3	47.9	46.1	53.0	54.4	48.2	50.7
TiO_2	2.2	0.96	0.65	0.64	0.58	0.55	0.11	2.87	0.88	2.32	2.52	0.94	0.67
Al_2O_3	15.3	14.6	12.3	12.6	13.8	13.8	2.7	14.8	10.5	14.6	13.3	12.2	15.9
Fe_2O_3	3.2	2.0	2.2	3.0	3.3	3.8	5.7	5.2	3.4	4.5	4.2	3.2	3.0
FeO	8.1	8.7	6.5	6.7	6.1	6.3	5.0	8.0	8.4	7.6	9.4	7.6	6.4
MnO	0.18	0.17	0.16	0.16	0.16	0.16	0.11	0.22	0.18	0.19	0.22	0.18	0.16
MgO	2.39	7.83	13.2	12.7	9.28	9.76	31.0	4.33	16.5	2.52	2.69	12.2	6.73
CaO	5.58	11.6	8.73	8.78	8.32	8.46	2.48	7.58	6.68	5.67	5.80	8.46	9.85
Na_2O	4.0	1.3	1.8	1.5	2.7	2.2	0.1	3.0	1.3	3.7	3.6	2.0	2.2
K_2O	2.89	0.28	1.50	1.29	1.56	1.40	0.04	1.88	0.67	2.55	2.58	1.14	1.48
H_2O	1.2	2.1	3.9	3.7	3.7	3.3	8.9	2.5	5.1	0.9	0.8	3.1	2.6
CO_2	0.1	0.9	0.1	0.2	0.0	0.0	1.0	0.7	0.2	0.1	0.2	0.4	0.1
P_2O_5	0.81	0.12	0.11	0.08	0.10	0.09	0.05	0.67	0.11	0.78	0.98	0.12	0.08
S	0.12	0.11	0.12	0.04	0.02	0.08	0.00	0.05	0.06	0.06	0.09	0.06	0.05

	761303	761402	761505	654706*	654708	654807*	654808	654809	654903	655404	655506	655607*	655608
SiO_2	55.5	53.2	53.9	49.9	50.2	53.2	53.2	53.9	53.7	45.3	51.3	53.8	54.9
TiO_2	1.70	2.54	2.14	1.03	1.17	2.75	2.55	2.54	2.58	4.00	3.22	2.61	2.22
Al_2O_3	17.3	14.3	15.5	14.7	15.4	14.0	14.3	14.2	14.5	16.5	14.0	14.3	15.5
Fe_2O_3	3.6	5.7	2.9	1.4	2.1	4.3	4.2	3.6	5.5	1.8	2.7	2.5	1.6
FeO	5.5	7.1	8.6	10.0	9.8	8.0	8.2	8.6	6.9	12.3	11.0	10.0	9.2
MnO	0.14	0.21	0.17	0.19	0.21	0.23	0.18	0.19	0.23	0.20	0.20	0.21	0.18
MgO	1.79	2.73	2.71	8.13	5.93	3.42	3.29	3.30	3.31	6.08	3.85	2.91	2.69
CaO	4.97	4.98	5.78	11.1	10.3	5.12	5.11	5.77	4.66	7.70	6.27	5.73	5.45
Na_2O	4.4	4.1	4.0	2.0	2.0	4.3	4.2	3.9	3.9	3.7	3.1	3.8	4.3
K_2O	3.14	2.82	2.47	0.46	0.89	2.90	2.44	2.49	2.45	1.07	2.61	2.78	2.72
H_2O	1.1	1.1	0.5	1.7	1.8	1.3	1.6	1.1	1.5	0.8	1.4	0.7	1.1
CO_2	0.1	0.0	0.1	0.0	0.0	0.5	0.2	0.0	0.1	0.2	0.0	0.1	0.1
P_2O_5	0.56	0.70	0.63	0.12	0.09	0.89	0.72	0.72	0.67	0.66	0.83	0.95	0.79
S	0.02	0.10	0.13	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND

NOTE: ND = not determined.
*These samples were obtained from chilled dyke margins; the remaining samples are either medium or coarse grained.

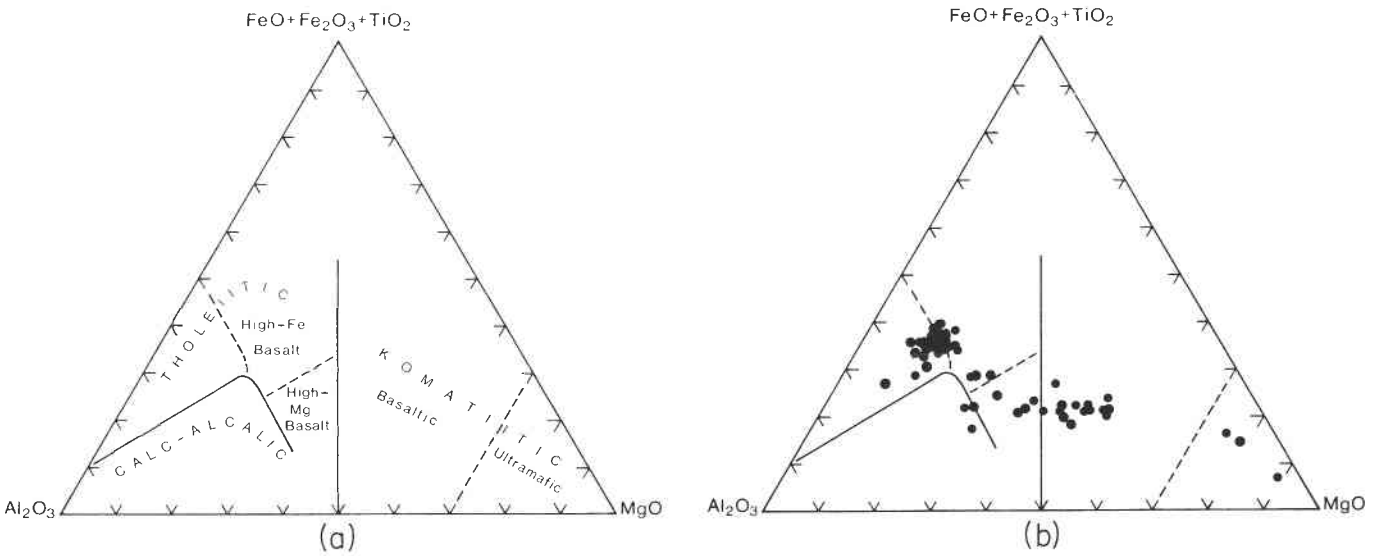


FIG. 3. (*a*) Jensen cation plot for classifying subalkalic rocks. (*b*) Analyses of Mistassini dyke material carried out at the University of Montreal (Table 1) and at the Geological Survey of Canada (Table 2).

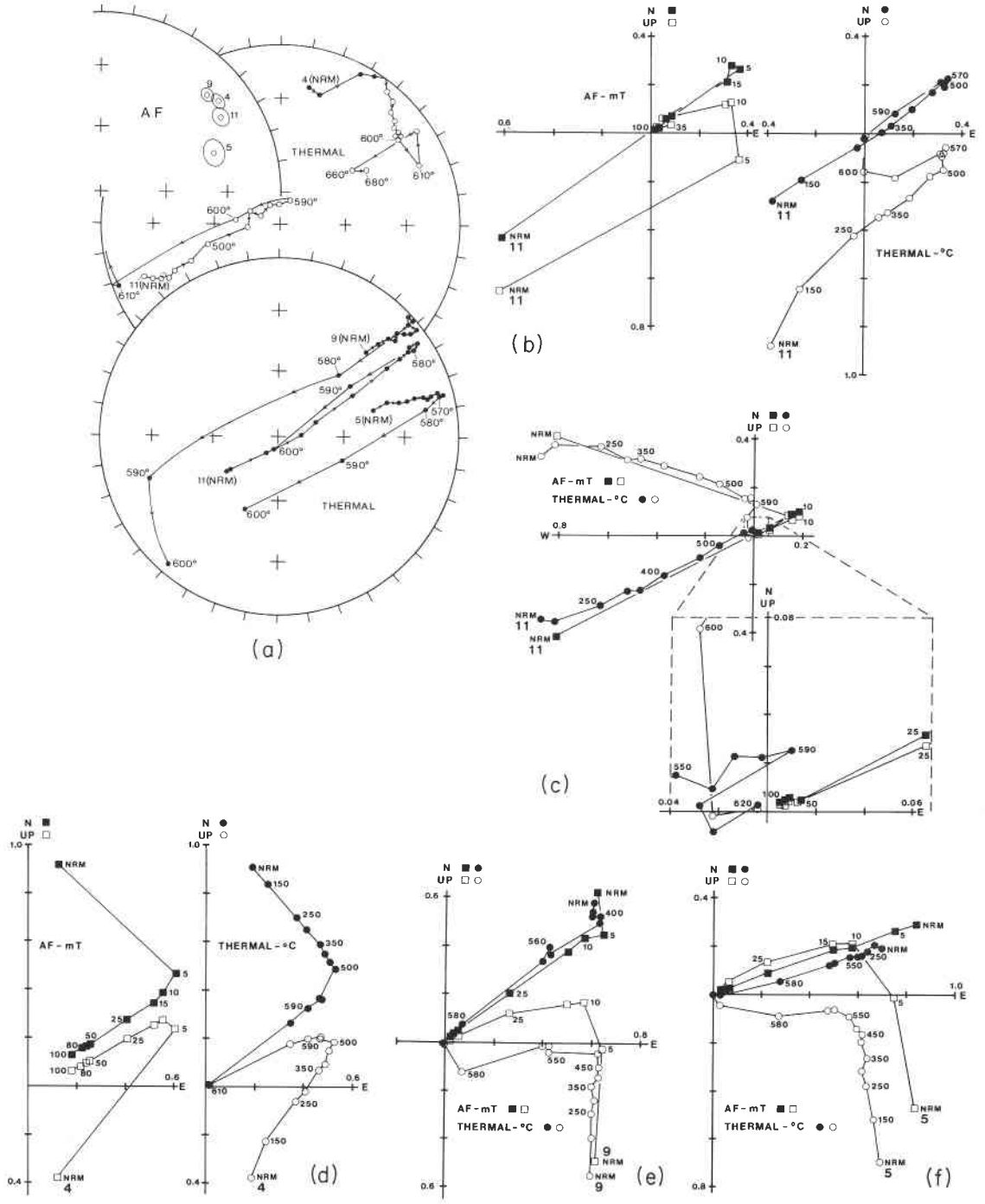


FIG. 4. (*a*) Results of step cleaning of some individual specimens from komatiitic sites 7604, -05, -09, and -11. Sites from which the specimens were taken are indicated by numbers followed by (NRM). Top left shows site mean directions that resulted from AF cleaning to 100 mT. (*b*–*f*) Zijderveld diagrams of AF and thermal cleaning for each of the five samples from four sites shown in (*a*). The number near the NRM indicates the site from which the sample was taken.

greater than 10%, whereas all the remaining 26 Montreal analyses indicate MgO contents of less than 7.2%. With the exception of the ultramafic komatiite, the Mistassini komatiitic suite has a significantly higher alkali content than comparable ocean-floor komatiitic basalt. This is not unexpected because of the continental environment of the Mistassini dykes. However, both K_2O and Na_2O contents are also notably higher than in environmentally similar komatiitic dyke suites, for example the Molson dykes (Ermanovics and Fahrig 1975; Scoates and Macek 1978). The K_2O content of the Mistassini komatiitic basalts is highly variable; for example, samples DJ1309 and EC0710, with MgO contents of 13%, have K_2O contents of 0.8 and 1.4%, and MW1204, with 12% MgO, contains only 0.2% K_2O. This K_2O variability may be the result of crustal

TABLE 3. Paleomagnetic data for Mistassini diabase dykes

Site	Cores	AF (mT)	Thermal (°C)	$D°$	$I°$	k	$\alpha_{95}°$	R
				Mistassini overprint				
7601	06	50						
	07	40		70.0	−1.0	—	—	
7603	01	60		31.0	12.0	—	—	
7604	01−08	100		51.0	−16.5	378	2.9	
7605	01−06	100		71.5	−34.5	204	4.7	
7609	01−07	50		48.0	−21.5	602	2.5	
7610	02	*						
	04	40		55.5	−34.5	—	—	
7611	01−07	60		58.0	−23.0	332	3.3	
7613	05	70						
	06	*		28.5	−09.0	—	—	
6555	01	60						
	02	35						
	03	20		72.0	−09.0	18	30	
6556	02	90						
	03	30						
	04	100						
	06	70		61.5	−20.0	21	20	
7603	02	100						
	05	70		235.0	19.0	—	—	R
7607	06	25		248.0	08.5	—	—	R
7609	03		590	251.5	26.0	—	—	R
Mean direction from 46 samples above				50.0	−17.5	20	9.6	
Pole position from mean direction				130.9°W, 12.9°S δ_m = 10.0°, δ_p = 5.2°				
				Very steeply down (primary?)				
7601	01	20						
	03	*						
	04	15						
	06	10						
	07	*		358.5	77.5	46	11.4	
7603	03	80		49.0	75.5	—	—	
7607	01	40	600	316.0	81.0	—	—	
	06	60						
7612	01−06	20		221.0	76.5	74	7.9	
7613	01	25						
	02	*		214.0	79.0	—	—	
7614	01	60						
	02		590	126.5	76.5	—	—	
6547	01	80		225.5	69.5	—	—	
6548	01	*						
	04	20						
	05	50		91.5	78.5	55	18	
6549	01	60		292.0	88.5	—	—	
6547	03	50		75.5	−73.5	—	—	R
7610	03	20		170.0	−75.5	—	—	R
Mean direction from 25 samples above				250.0	88.0	29	5.5	
Mean pole from mean direction				80.0°W, 50.0°N δ_m = 11.0°, δ_p = 11.0°				
				NE-down and reversed				
7603	04	60		255.0	−37.5	—	—	R
7607	02	100						
	04	35						
	06		640	226.0	−25.0	29	23	R
7610	06		620					
	07		600	236.0	−33.0	—	—	R
7601	01	35						
	03	35						
	04	70		52.0	54.0	46	13.8	
7607	02		560	25.5	64.5	—	—	
7608	03	100		35.5	24.5	—	—	
7610	06		580					
	07		580	63.5	37.5	—	—	

TABLE 3. (*Concluded*)

Site	Cores	AF (mT)	Thermal (°C)	$D°$	$I°$	k	$\alpha_{95}°$	R
7615	01	35						
	02	25						
	03	20						
	04	30		49.5	51.0	27	18	
6555	05	40		50.0	38.5	—	—	
6556	01		590	42.5	30.0	—	—	
Mean direction from 19 samples above				50.9	42.0			
Mean pole from mean direction				146.8°W, 42.1°S δ_m = 12.1°, δ_p = 7.3°				

NOTES: AF = peak alternating field; an asterisk in this column indicates that the result for the sample (core) is a mean of data for two or more specimens from that sample that were treated differently: either they had different AF end points or a mixture of results from AF and thermal treatment are used; Thermal = cleaning temperature; D = declination; I = inclination; k = Fisher's estimate of precision; α_{95} = semi-axis of the circle of confidence at $P = 0.05$; R is arbitrarily used to emphasize in the case of the "Mistassini overprint" and "NE-down" that the sample has a southwest-up or -down rather than the more common antipodal northeast-up or -down direction; in the case of very steep components R refers to a very steeply up direction; δ_m, δ_p are semi-axes of ellipse of confidence at $P = 0.05$.

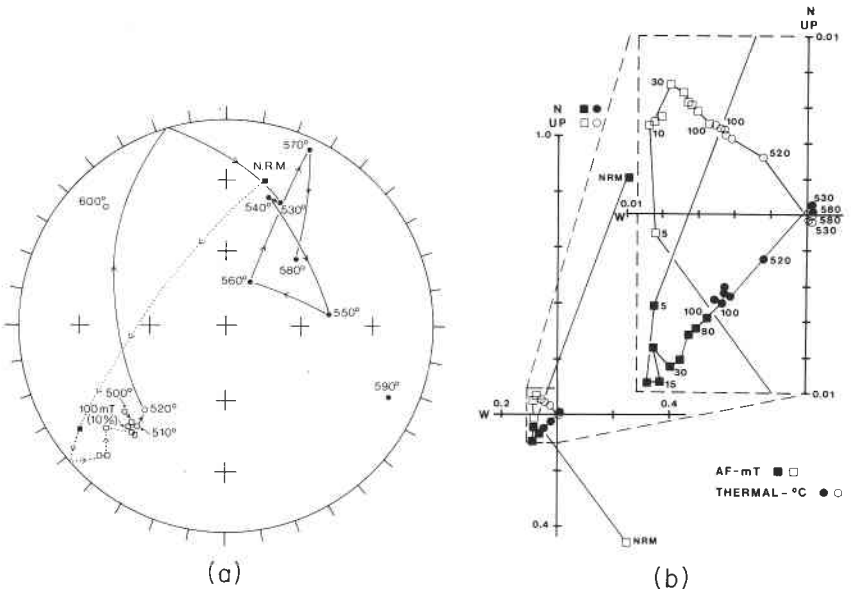


FIG. 5. (*a*) Result of AF cleaning of specimens from core 2 of site 7607 to 100 mT followed by thermal step cleaning. (*b*) Zijderveld diagram for specimen shown in (*a*). The number near NRM is the sample number.

contamination. The presence of spectacular zones of large plagioclase phenocrysts within some of the tholeiitic dykes speaks of crystal fractionation, which may well have taken place in a magma chamber within the crust. The abundance of TiO_2 in the MgO-rich members of Mistassini swarm is notably lower than in the tholeiitic suite and is in the range commonly found in komatiitic suites elsewhere, for example in the Molson dykes. Cr and Ni in the komatiitic suite correlate well with MgO content, suggesting fractionation of olivine and pyroxene.

Some analyses were obtained from chilled dyke margins (Table 2) but are too few to allow a statistically valid comparison between the chemistry of dyke margins and that of the central zones.

Paleomagnetism

Two and, in many cases, three specimens from each of 114 solar-oriented samples were alternating field (AF) demagnetized in 15 steps to 100 mT, and at least two specimens from each site were thermally step demagnetized to 580°C and in some cases to 680°C. The Mistassini dykes do not outcrop well, so in only a few localities could samples be obtained from near dyke margins. In all other cases the samples were obtained as regularly as possible across the dyke. In some cases, to spread out sampling, it was necessary to sample two or more outcrops along the strike of the dyke. Within these limitations no correlation could be observed between position of samples relative to dyke margins and paleomagnetic results (Table 3).

Mistassini overprint

Drill cores were collected for paleomagnetic study at five sites representing the komatiitic suite, and results from four of these are described first because much of our interpretation of the paleomagnetism of the Mistassini dykes hinges on results from these sites. Material at four of the sites, 7604, -05, -09, and -11, is komatiitic basalt, and at the fifth site, 7607, ultra-

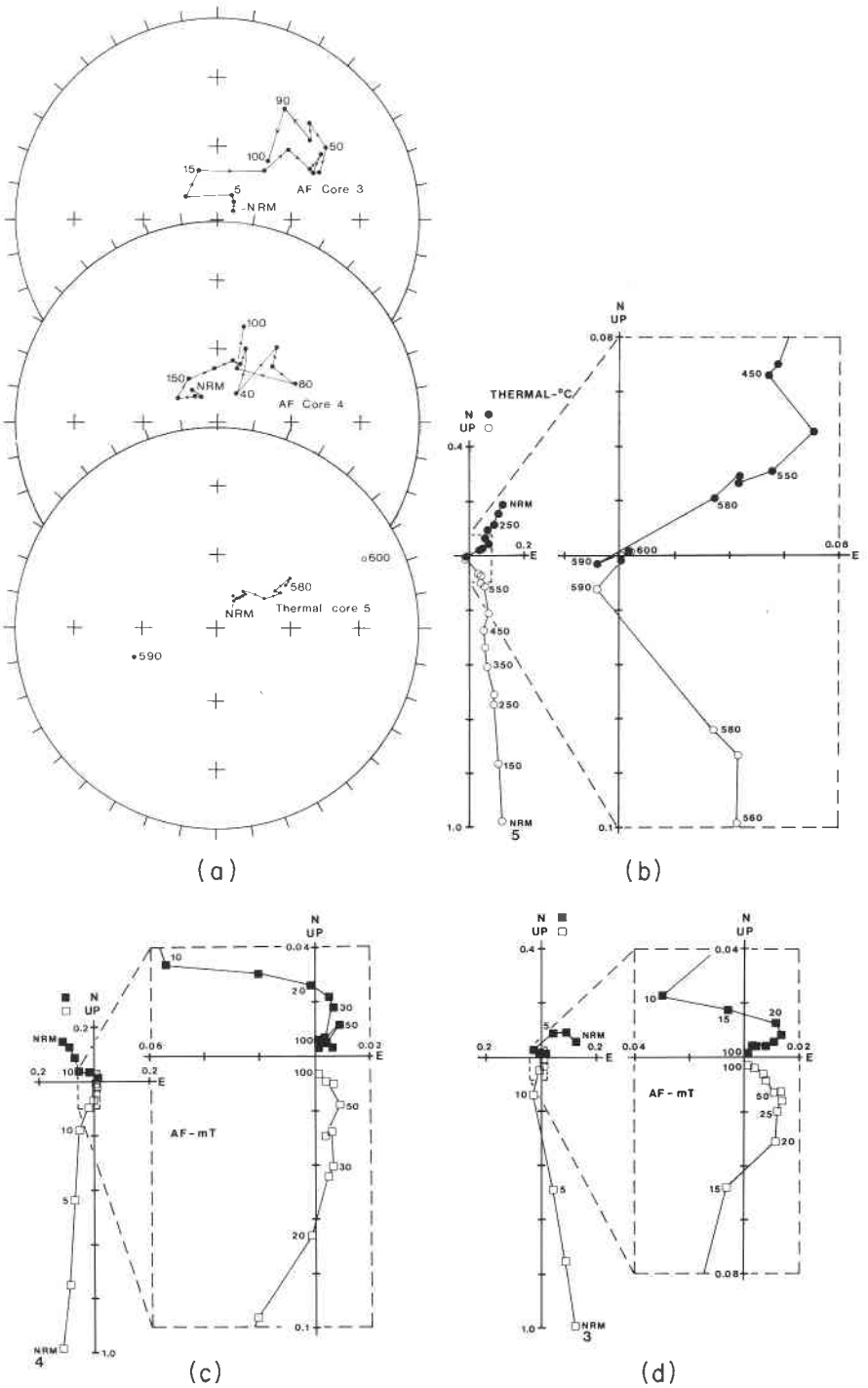


FIG. 6. (*a*) Thermal and AF step cleaning of specimens from samples 3, 4, and 5 from site 7601. (*b*–*d*) Zijderveld diagrams for cores 3, 4, and 5 above.

mafic komatiite. All the known komatiitic dykes occur within 45 km of Mistassini Lake.

The magnetization direction of all 28 cores (56 specimens) of the four komatiitic basalt sites moved to a stable northeast-up direction in response to step AF demagnetization in fields up to 100 mT (Fig. 4). With the exception of one sample, all showed evidence of only this component except at low fields, i.e., below 10 mT. The one exception had a natural remanent magnetization (NRM) intensity 10^2 times that of the group mean, but it too exhibited only the group direction in fields above 25 mT. It is thought that this core was affected by a lightning strike.

At least two specimens for each of the komatiitic basalt sites were thermally demagnetized. The northeast component is apparent in all except the apparently lightning-struck sample. In two samples only the northeast component is present, and in one of these it is clear that the northeast-up component resides in both magnetite and hematite, as this direction is maintained up to 680°C (see specimen labelled 4, Fig. 4). The other specimens exhibited a shift towards, or to, the northeast-up

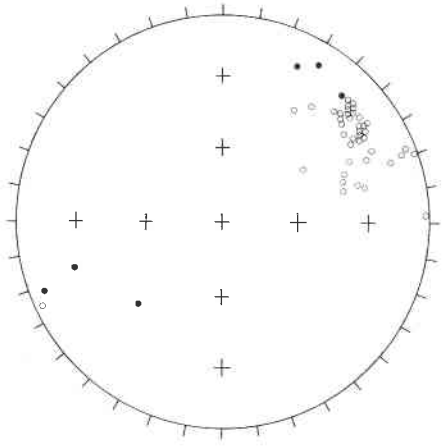


FIG. 7. Cleaned directions of 46 samples exhibiting northeast-up (and reversed) overprint component.

disturbance, the age of this component is in the range 1400−1500 Ma. If, as previously argued, the dykes are Early Proterozoic in age, this northeast-up component is an overprint possibly related to the Elsonian Disturbance (Stockwell 1982). Supportive evidence is the closeness of the four sites to Elsonian intrusions across the Grenville Front and the susceptibility of these rather mafic rocks to alteration. There is no clear petrographic evidence of alteration (deuteric or otherwise) based on position relative to the Grenville Front. However, this does not preclude the possibility of the imposition of a later, regionally variable chemical remanent magnetization (CRM) component. Whole-rock K−Ar ages of 1230 ± 45 and 1265 ± 165 Ma from chilled material of sites 48 and 47 are in reasonable agreement with an Elsonian overprint because whole-rock K−Ar ages are generally 10−15% lower than ages of primary crystallization or later thermal metamorphism.

The northeast-up component (Mistassini overprint) also occurs in six of the tholeiitic sites. In none of these sites (7601, -03, -10, -13, 6555, and -56) does this component occur alone.

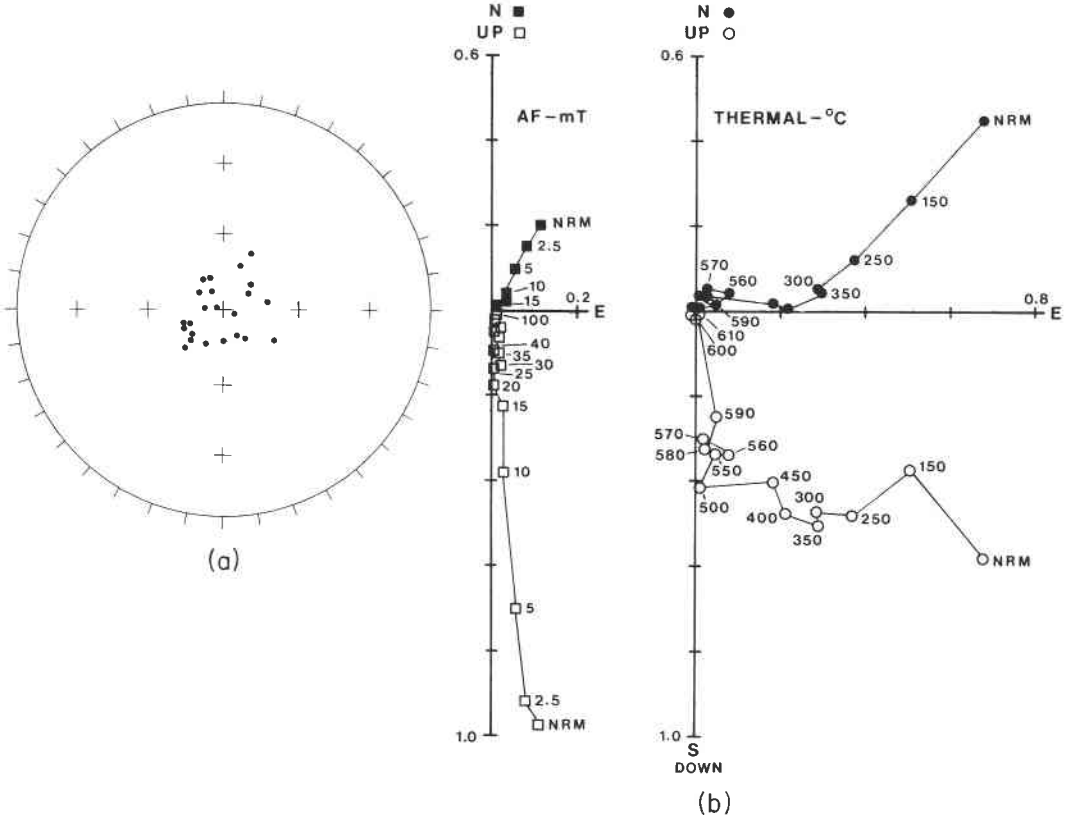


FIG. 8. (*a*) Cleaned directions of 25 samples from 11 sites giving very steeply down (and reversed) component. (*b*) Zijderveld diagram for two specimens of core 01, site 14.

direction, but in most cases they shifted back towards what is assumed to be a reversed direction. This move towards a reversed direction takes place only above 570°C, indicating that the reversed component resides in hematite. We evidently have a dominant northeast-up component residing in both magnetite and hematite and a minor hematite component that may be reversed to this in four of five komatiitic sites (Fig 4).

The pole position of the four komatiitic basalt sites is about 130°W, 11°S. On the basis of the Berger and York (1980) polar wandering path and barring the possibility of tectonic

Furthermore, a southwest, presumably reversed component occurs, as previously noted, in four komatiitic sites and in two tholeiitic sites (7603, 7607). In total, the northeast-up (and reversed) component was isolated in 46 samples from 11 sites.

Steeply down (primary?) component

A stable, steeply down component is present alone in the material collected at three sites: 7612, 7614, and 6549.

The magnetization of site 12 remained very steeply south down during step AF cleaning up to 60 mT and thermal clean-

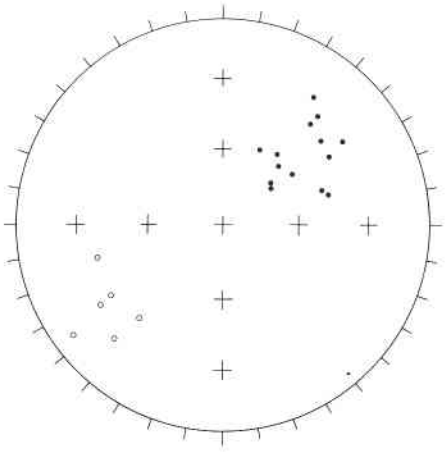


FIG. 9. Northeast-down (and reversed) directions from 19 specimens of 10 sites.

stable, very steeply down component in response to both AF and thermal step cleaning, and sites 6547 and 7610 each contain one core that has a stable, very steeply up component.

The pole position for the steeply down (and reversed) component is 080°W, 50°N. This pole position is east of that expected for rocks magnetized >2000 Ma ago; however, a northeast-up Elsonian overprint would cause an apparent eastward shift of the pole and would also explain the rather large ellipse of confidence for this pole.

On the other hand, it is conceivable that this component is also an overprint. If so, its pole position might suggest very late Hudsonian overprinting. This seems less likely, as the dykes farther from the front have been found less likely to contain the northeast-up (overprint?) component.

Northeast-down and reversed component

Three sites (7603, -07, and -10) yielded six cores with stable, gently to moderately steeply southwest-up directions. In the case of site 03, cores 4 and 5 were gently southwest up

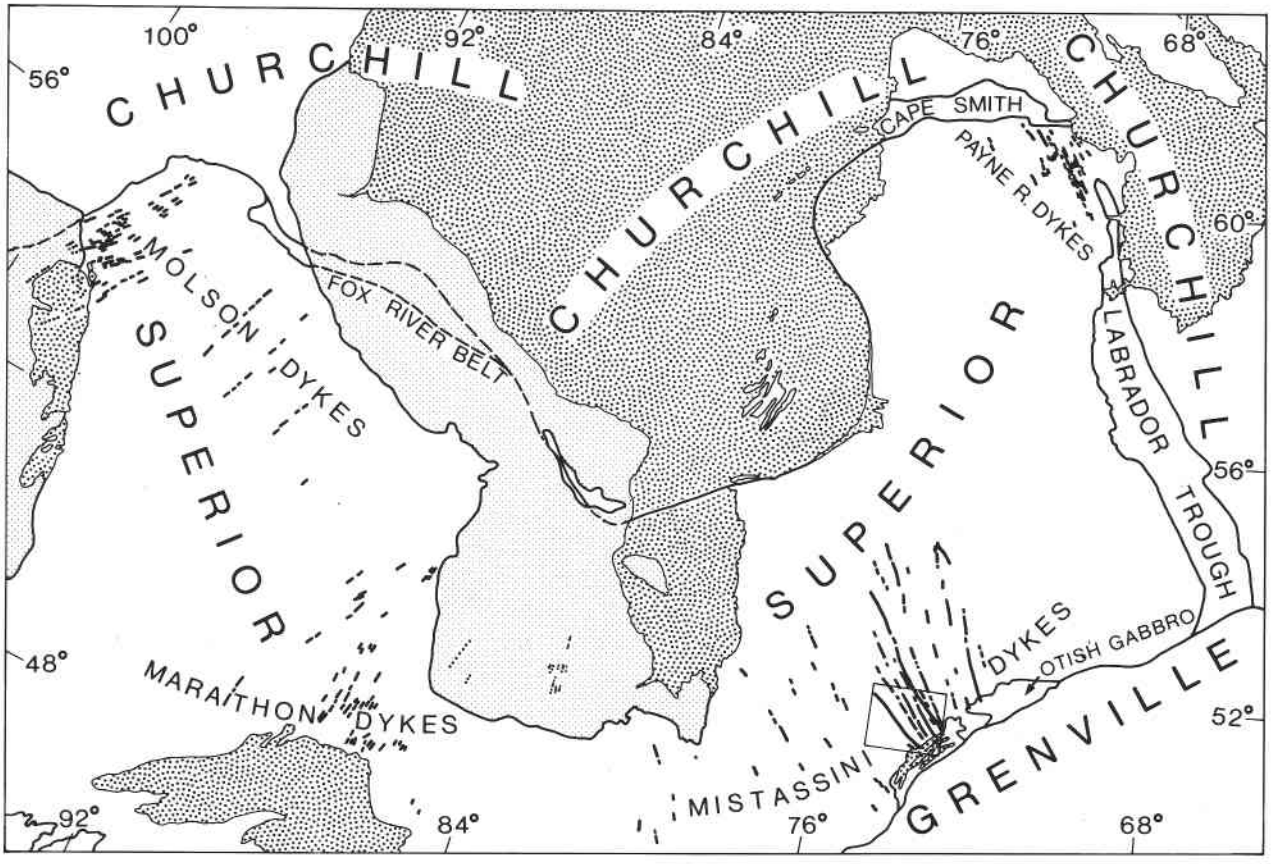


FIG. 10. Early Aphebian to early middle Aphebian basic extrusive rocks peripheral to the Superior Province and some possibly related continental diabase dyke swarms.

ing to 570°C. Samples from site 14 also retained only a steeply down component, in some cases after step cleaning to 80 mT with a loss of 98% of NRM intensity. Thermal cleaning also revealed only the steeply down component with a sharp decline in intensity from 25% of NRM intensity to less than 2% between 590 and 600°C. Only two samples, one of which indicated a stable, very steeply down magnetization, were collected at site 49.

Six additional sites (7501, -03, -07, -13, 6547, and -48) have one to three cores exhibiting unequivocal evidence of a

and moderately southwest up. Thermal cleaning of specimens from these cores duplicated the AF result. In site 07, five cores shifted to a southwest direction, but only cores 2, 4, and 6 had a negative inclination. In site 10, the core directions shifted to a northeast-up direction during step AF cleaning, but in response to thermal cleaning specimens from cores 4, 6, and 7 after moving to a shallow northeast-down direction moved above 580°C to a southwest-up direction. It occurred to us that these southwest-up directions might represent an uncompleted move to a very steeply up direction, but the behaviour, in

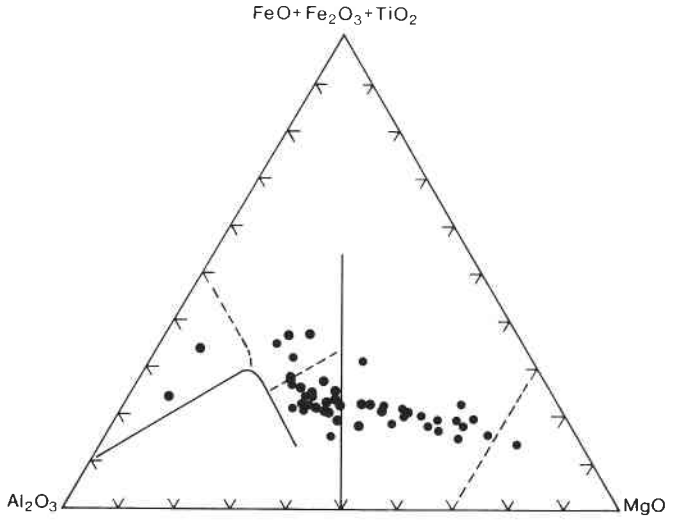


FIG. 11. Jensen plot of Molson dyke analyses (Ermanovics and Fahrig 1975; Scoates and Macek 1978).

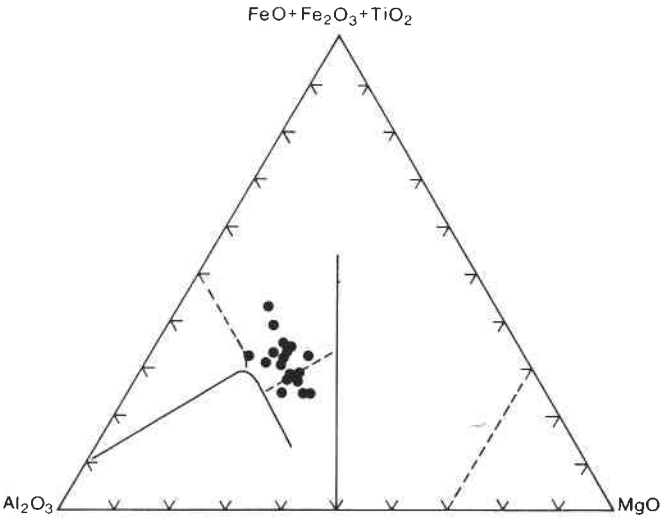


FIG. 13. Jensen plot of chemical analyses of Marathon dykes. Whole-rock K–Ar ages = 1550–1815 Ma.

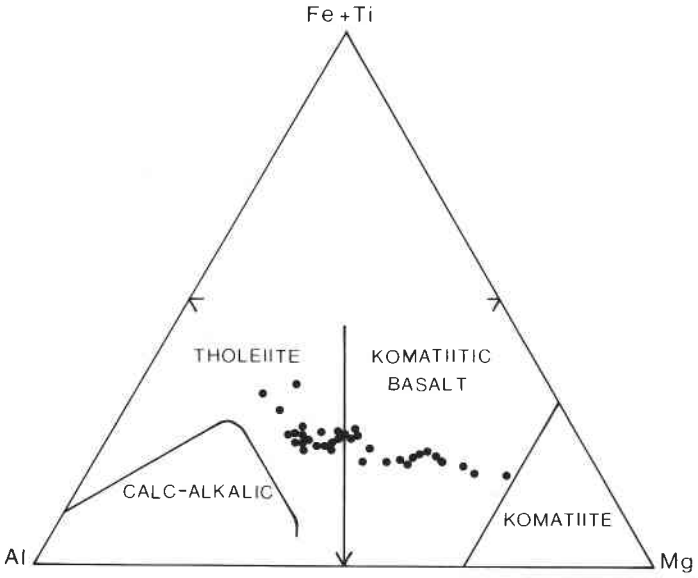


FIG. 12. Jensen plot of Fox River Belt analyses (Scoates 1981).

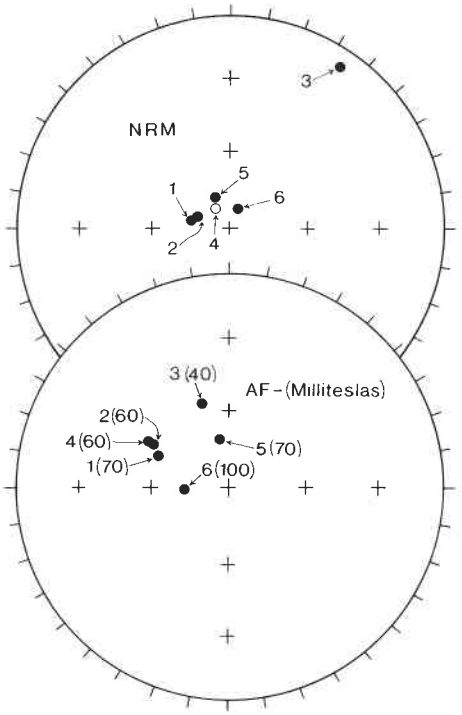


FIG. 14. Results of detailed AF cleaning of samples from a single site on a Marathon dyke located near Geco mine, Manitouwadge area.

particular of core 2 of site 7 (Fig. 5), seems to contradict this. The core has an extremely stable southwest-up direction in response to AF cleaning to 100 mT. Three specimens from this core, after having been AF step cleaned to 100 mT, were subjected to thermal step cleaning. All three exhibited a sharp drop in initial NRM intensity from about 10% to <1% after cleaning at 530°C. This was accompanied by a change to a northeast-down (presumably reversed) direction (Fig. 5). Because of this behaviour, we were convinced that a southwest-up (and reversed) component had been isolated.

The northeast-down component may have been isolated in five sites besides sites 7607 and -10, as mentioned above. These are sites 7601, -08, -15, 6555, and -56). These northeast directions were of particular concern because of the possibility that they represent an unresolved mixture of a very steeply down component with the proposed northeast-up overprint. Site 7601 is a case in point. Three of six cores yielded gently up northeast end points, but cores 7603 and 7604 exhibited a very stable, northeast-down component with a steady decrease in initial NRM intensity to <1% at 100 mT (Fig. 6). Thermal work also supported the establishment of this northeast-down direction. These results, however, do not preclude the possibility that this northeast-down direction is a mixture of the northeast-up and very steeply down components.

Discussion of paleomagnetic results

Extensive laboratory work on 114 independently solar-oriented samples from 20 sites on the Mistassini swarm has indicated a complex history of magnetization. The material

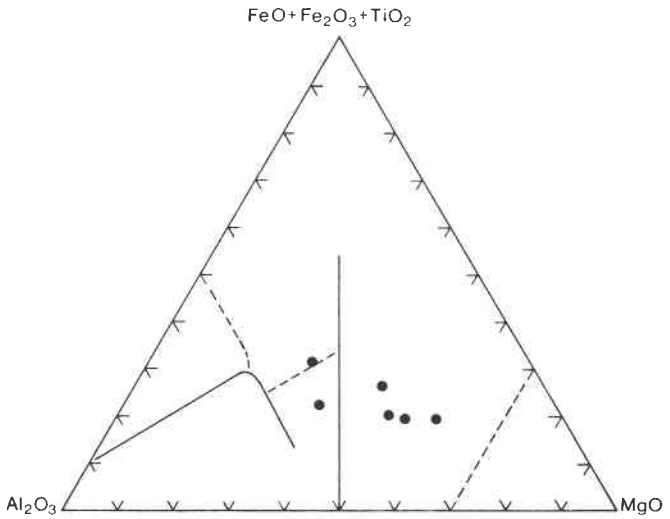


FIG. 15. Jensen plot of chemical analyses of Cape Smith volcanic samples (Schwarz and Fujiwara 1977). Sm–Nd age = 1871 ± 75 Ma (Zindler 1982).

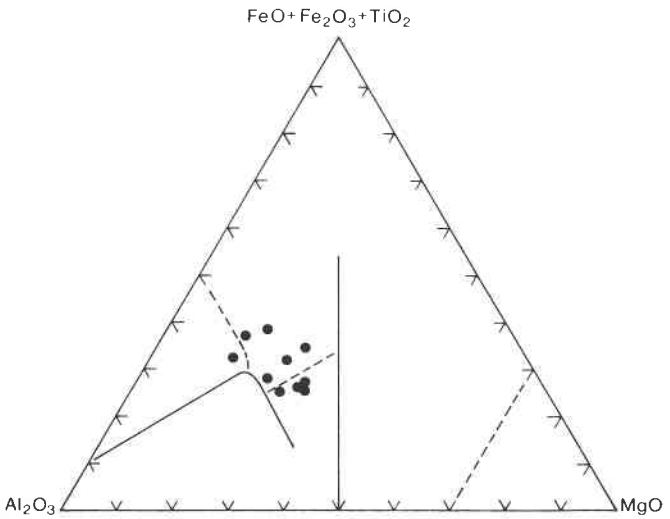


FIG. 16. Jensen plot of chemical analyses of samples from the Payne River dykes, New Quebec. K–Ar dates on chilled whole-rock samples = 1875 ± 240 and 1790 ± 240 Ma.

from four of five komatiitic dyke sites yielded, in response to AF cleaning, a single northeast-up component and, with thermal cleaning at high temperature, a component apparently reversed to this, which is held by hematite. This northeast-up or reversed component was also identified in nine sites from dykes of the tholeiitic suite, but in these sites other components are also present. The pole position for the 46 cores giving the northeast-up direction is 131°W, 13°S. This pole suggests magnetization with an age of 1400–1500 Ma, in good agreement with two chilled whole-rock K–Ar ages of about 1250 Ma, considering that whole-rock K–Ar ages of chilled mafic material are generally 10–15% less than ages of intrusion or remagnetization. Two other K–Ar ages have indicated a much greater age for the intrusions, so we suggest that the northeast-up (and reversed) component (Fig. 7) is an overprint related to the Elsonian Disturbance. It is likely that skeletal opaques of the tholeiites contain the older, steeply down remanence. Probably the second generation of magnetite in the

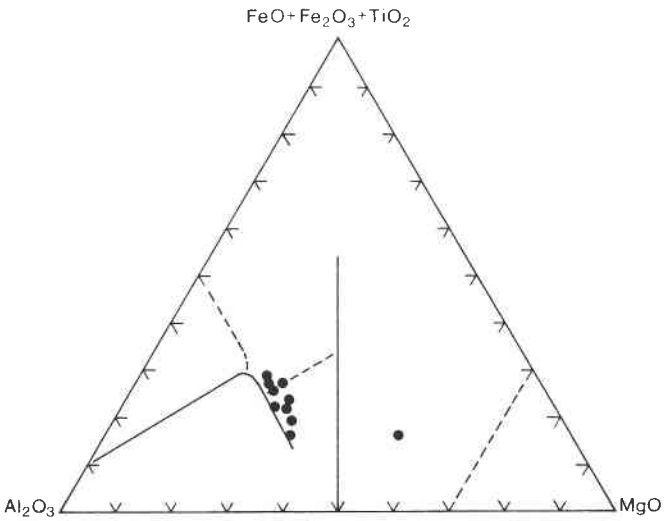


FIG. 17. Jensen plot of chemical analyses of samples from the Otish gabbro.

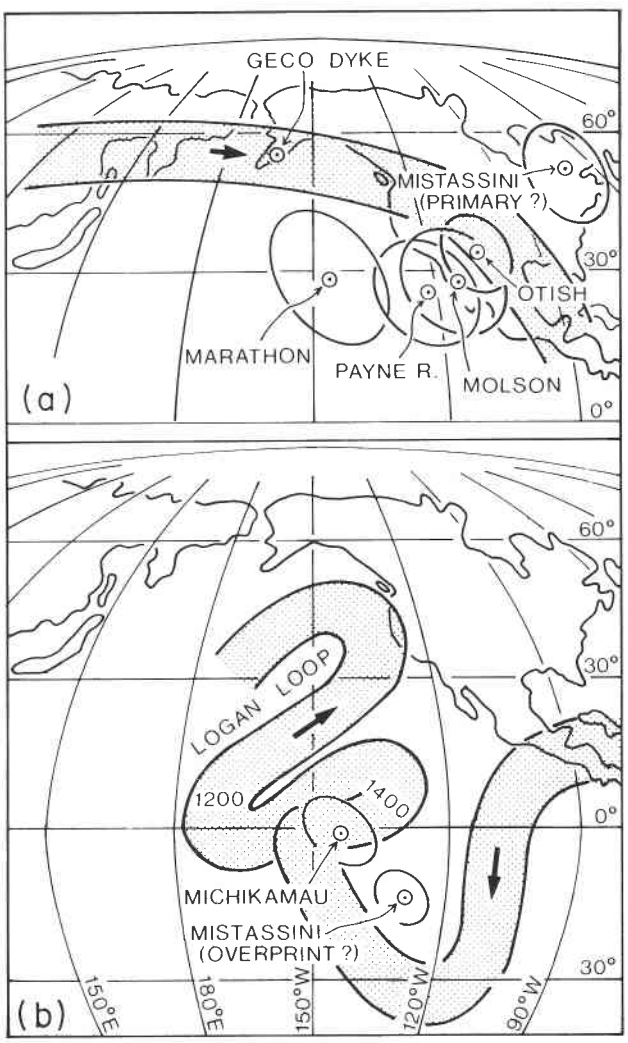


FIG. 18. (*a*) position of the Mistassini overprint relative to the Helikian polar wandering curve (chiefly from Berger and York 1980). The Michikamau pole (Murthy *et al.* 1968), which represents the Elsonian Disturbance, is shown for comparison. (*b*) Poles for various dyke swarms, the Otish gabbro, and a single Marathon (Geco) dyke.

TABLE 4. Paleomagnetic data for Payne River dykes

	NRM						Cleaned				
Site	N	n	$D°$	$I°$	k	$\alpha_{95}°$	AF (mT)	$D°$	$I°$	k	$\alpha_{95}°$
6568	6	6	253	77	94	7	50–90	224	56	80	8
6570	5	5	282	78	43	12	20–60	211	66	21	17
6572	6	5	263	76	53	11	30–70	260	60	44	12
6573	3	3	289	73	73	15	25–50	284	52	148	10
Mean direction of 19 cores above (cleaned)								242	62	19	8
Mean pole position								118.0°E, 25.5°N δ_m = 12°, δ_p = 9.5°			

NOTES: N = number of cores collected at a site; n = number after eliminating unstable cores; D = declination; I = inclination; k = Fisher's estimate of precision; α_{95} = semi-axis of the circle of confidence at $P = 0.05$; AF = range of peak alternating fields established as the end points for samples of that site; δ_m , δ_p = semi-axes of ellipse of confidence at $P = 0.05$.

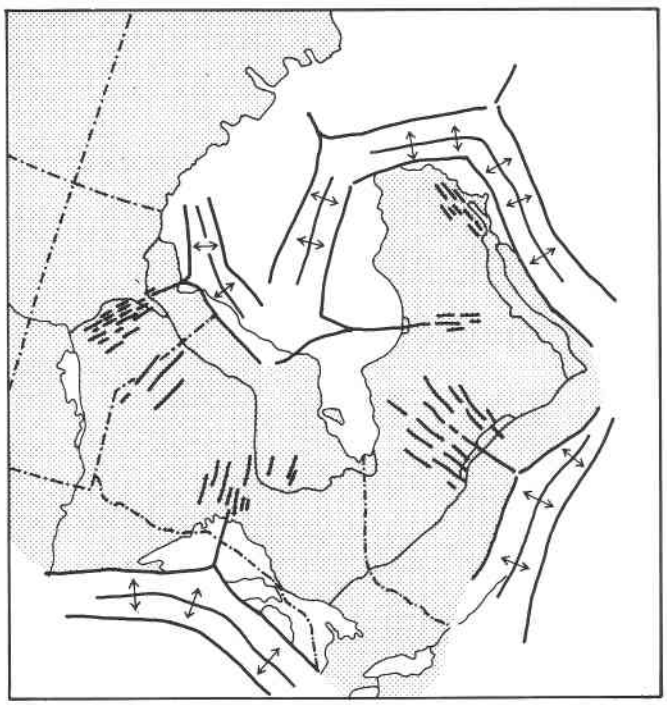


FIG. 19. The probable tectonic relationship between continental diabase dyke swarms and ocean opening. Dykes appear in two contexts: most swarms appear to be products of failed arm tensional tectonics; less commonly swarms are preserved parallel to paleocontinental edges where they formed during an early stage of ocean spreading. Note that we have included the Richmond Gulf graben (Chandler and Schwarz 1980).

komatiites (and the hematite) contains the strong northeast-up and reversed component.

Twenty-three specimens yielded very steeply down components, and two a very steeply up component. These specimens were derived from 11 sites, and very detailed AF and thermal work indicates that this is largely an independent component (Fig 8). The pole position, 080°W, 50°N, is somewhat to the east of that expected from a primary component >2000 Ma in age. However, there is a distinct possibility that some of these specimens contain elements of the above-described overprint, which had the effect of shifting the pole for the very steep component eastward, away from the polar wandering curve. The apparent northeast-down component isolated in some specimens may be evidence of streaking between the very steeply down and the northeast-up direction.

A third group of 19 samples from 10 sites yielded northeast-down and reversed, southwest-up directions (Fig. 9). Their pole position at 147°W, 42°S is a considerable distance away from any part of the polar wandering curve for the Superior Province. Although results from some specimens give no evidence of mixed components, the southwest-up direction can be explained in terms of an interplay of the two components described above if a complex history of magnetization is postulated. The southwest-up component would have to represent a very steeply up (primary?) component overprinted by the southwest-down component. The fact that all components lie in a single vertical plane suggests that the southwest-up–northeast-down components are indeed a mixture of the very steep (primary?) and northeast-up and reversed (overprint?).

The broader significance of Mistassini and other basic continental dyke swarms

The Mistassini dyke swarm is one of a group of basic dyke swarms that occurs around the periphery of the Superior Province (Fig. 10). These swarms are early or possibly early middle Aphebian (2000–2200 Ma) in age, and by examining these swarms and their tectonic setting it may be possible to say something about the tectonic significance of all the world's great continental diabase dyke swarms (Fahrig *et al.* 1984).

The Molson dykes of northeastern Manitoba and northwestern Ontario were first described in the last century. Very preliminary Molson paleomagnetics were published by Fahrig *et al.* (1965), and because the Molson dykes were known to contain ultramafic representatives, their systematic study was begun by Scoates in 1969 as part of an ultramafic rock project. Ermanovics and Fahrig (1975) published a petrochemical–paleomagnetic report on the dykes that suggested the dykes were intruded between 1800 and 2000 Ma ago. Scoates and Macek (1978) specifically noted that the more basic dykes of the swarm form a komatiitic suite (Fig. 11). The Molson dykes strike directly towards the Fox River Belt, which also contains a komatiitic suite (Fig. 12) (Scoates 1981). It seemed reasonable at least on chemical grounds to suggest a consanguineous relationship among the mafic rocks of the Fox River Belt, those of the Thompson Belt, and the Molson dykes.

The Marathon dykes strike north–south along the north rim of Lake Superior. Very little published work is available on

these dykes, but very preliminary paleomagnetic and chemical data (Fig. 13) on them were published by Fahrig *et al.* (1965). Recent detailed paleomagnetic results on a single dyke site near Geco mine are shown in Fig. 14. This dyke seems to contain only one steeply down component, corroborating early work. A biotite concentrate from this site yielded a K–Ar age of 1705 Ma, and five other whole-rock K–Ar ages from the swarm ranged from 1550 to 1950 Ma. These K–Ar ages suggest an age of intrusion over 2000 Ma. The Marathon dykes have compositions lying across the high-magnesium–high-iron tholeiitic basalt boundary.

The Cape Smith Belt forms the northeast boundary of the Superior Province, and from this belt high-magnesium basalts were reported by Wilson *et al.* (1969). Schwarz and Fujiwara (1977) were the first to indicate that some of the basic rocks of the belt had consolidated from komatiitic magma (Fig. 15). Schimann (1978) reported a Rb–Sr age of 2150 Ma from volcanics of the belt, and Zindler (1982) reported a Sm–Nd isochron age of 1871 ± 75 Ma from a sill and differentiated flow. The only known dyke swarm that might be related to the Cape Smith Belt is the Payne River swarm, which is exactly on strike with the eugeosyncline of the Labrador Trough and its great accumulation of mafic and ultramafic rocks. Paleomagnetic data are available from four sites on the Payne River dykes (Table 4), and chemically these dykes lie across the high-magnesium–high-iron tholeiite boundary (Fig. 16). Two chilled margin samples yielded whole-rock K–Ar ages of 1875 and 1790 Ma, suggesting an age of intrusion greater than 2000 Ma.

The Otish gabbro, which intrudes the Otish Mountain Group just east of the Mistassini dykes, comprises three main sills with minor subsidiary sills and dykes. These have a high-magnesium tholeiite composition (Fig. 17) except for a sample from finely laminated basal material, which is presumably an ultramafic cumulate. The Otish gabbro has a very steep, westerly magnetization (Fahrig and Chown 1973). A whole-rock K–Ar age of 1591 ± 37 Ma (Stevens *et al.* 1982, p. 43) is probably much less than the age of gabbro intrusion. Figure 18 shows the present state of knowledge regarding the paleomagnetism of the various dyke swarms, the Otish gabbro, and the virtual pole for the Geco (Marathon) dyke.

In summary, the Superior Province is intersected by groups of basic dykes that are of similar, although certainly not identical, age, i.e., probably 2000–2200 Ma. It is also surrounded by basic volcanic belts of possibly similar age, and some of the dyke swarms and volcanic belts contain komatiitic suites. A likely scenario relating the dykes and volcanic belts is shown in Fig. 19. It is suggested that in most cases the dyke swarms represent failed arms of spreading (triple?) points. The present volcanic belts would represent ocean openings and subsequent closing. An initial test of this hypothesis should be an attempt to relate dyke swarms throughout geological history to ocean openings of appropriate age.

Payne River dykes are probably an extension of an early opening phase that resulted in the development of the Labrador Trough. Swarms of this type would generally be destroyed during ocean closing.

A corollary of this hypothesis is that continental diabase dyke swarms generally will not extend from continental plates beyond peripheral, younger fold belts into what once constituted continuations of those continental plates. Dyke swarms, except for their paleomagnetic signature, will usually be of limited use in reconstructing supercontinents. There will be exceptions to this situation. In some instances dyke swarms generated during an early spreading episode may be bisected by a second, unrelated spreading event, in which case the dyke swarm may appear in recognizable form in two plate segments separated by an ocean or a supracrustal fold belt. Unravelling tectonic history will require the careful determination of dyke ages and the relating of these ages to ages of spreading episodes. Furthermore, if a spreading episode aborts, so that no ocean opening takes place, evidence for the episode may be difficult to find and indeed may consist of only the dyke swarms themselves. The number and extent of continental dyke swarms during a geological epoch may be a measure of the number and vigour of spreading events.

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